

HOHN, Germany

WMO: 100380

Lat: 54.316N

Lon: 9.539E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-6.9	-11.5	1.4	-7.5	-9.2	1.7	-5.6	13.8	6.5	12.2	5.9	2.5	110	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	28.3	19.1	26.1	18.4	24.1	17.5	20.2	26.3	19.2	24.6	18.2	22.9	3.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	22.5	17.1	12.2	21.5	16.1	11.5	20.2	58.0	26.8	54.4	24.5	51.3	23.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.3	8.2	-13.0	32.0	3.2	1.8	-15.2	33.3	-17.1	34.4	-18.9	35.4	-21.2	36.7		
			-13.2	22.4	3.1	1.0	-15.4	23.1	-17.2	23.7	-19.0	24.2	-21.2	24.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	1.4	1.9	4.1	8.2	11.9	14.9	17.4	17.3	14.0	9.8	5.4	2.4
	DBStd	6.73	4.28	4.02	3.55	3.52	3.30	2.97	3.04	2.77	2.72	3.37	3.51	4.32
	HDD10.0	1200	268	228	184	76	18	1	0	0	2	46	143	236
	HDD18.3	3438	526	461	441	304	202	112	56	55	134	265	389	493
	CDD10.0	864	0	0	1	22	76	147	230	225	121	39	3	1
	CDD18.3	61	0	0	0	0	2	8	27	22	2	0	0	0
	CDH23.3	608	0	0	0	4	19	87	279	197	21	1	0	0
	CDH26.7	149	0	0	0	0	2	18	77	51	2	0	0	0
(14) Wind	WSAvg	3.8	4.3	4.4	4.5	4.0	3.8	3.7	3.5	3.1	3.2	3.4	3.6	4.0
(15) Precipitation	PrecAvg	851	74	55	50	43	57	72	93	87	80	79	80	82
	PrecMax	1077	156	145	115	111	102	141	178	215	171	186	183	192
	PrecMin	570	1	8	7	6	15	1	34	29	17	25	19	27
	PrecStd	123	40	28	28	23	24	34	43	43	39	32	44	43
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.7	16.2	23.2	25.7	29.1	31.9	30.8	26.0	20.0	14.5	12.1
		MCWB	9.2	8.3	10.4	14.7	17.0	19.4	20.1	19.5	18.6	15.5	12.6	10.5
	2%	DB	9.2	9.8	13.0	19.5	22.9	25.9	28.8	28.0	23.0	17.2	12.9	10.7
		MCWB	8.0	7.6	8.9	12.4	15.5	18.1	19.3	19.0	17.1	14.2	11.3	9.4
	5%	DB	8.0	8.2	10.8	17.0	20.8	23.2	26.5	25.4	20.8	15.9	11.4	9.1
		MCWB	7.0	6.8	8.1	11.5	14.5	16.9	18.6	18.3	16.0	13.3	9.8	8.1
	10%	DB	6.9	7.1	9.2	14.5	18.7	21.0	24.1	23.1	18.8	14.7	10.2	8.0
		MCWB	6.0	5.9	7.0	10.3	13.4	15.7	17.7	17.3	15.0	12.6	8.9	7.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.2	11.3	15.0	18.1	21.0	21.5	21.2	19.4	16.2	13.2	11.1
		MCDB	10.5	10.7	14.8	21.7	23.2	26.7	27.9	27.2	24.4	19.1	14.4	12.2
	2%	WB	8.1	7.8	9.7	13.1	16.4	19.0	20.4	19.8	17.7	14.6	11.5	9.5
		MCDB	8.9	9.0	12.0	18.1	21.1	24.4	27.0	25.2	21.5	16.8	12.8	10.3
	5%	WB	7.0	6.9	8.5	11.8	15.0	17.5	19.4	18.7	16.5	13.7	10.2	8.2
		MCDB	7.7	8.0	10.5	16.0	19.3	21.9	25.1	23.8	19.8	15.5	11.2	8.9
	10%	WB	5.9	5.9	7.2	10.6	13.8	16.2	18.2	17.8	15.6	12.8	9.1	7.0
		MCDB	6.7	7.0	8.7	14.2	17.9	20.0	22.3	22.3	18.2	14.5	10.0	7.7
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	7.1	9.4	10.1	9.5	9.8	9.7	8.5	7.0	5.2	4.5
		MCDBR	4.6	6.2	9.9	13.6	14.0	13.7	14.1	13.6	11.4	8.2	5.4	4.4
		MCWBR	4.2	4.8	6.7	7.9	7.9	7.6	6.6	6.6	6.6	5.7	4.5	4.2
	5% WB	MCDWR	4.3	5.4	8.7	11.8	11.9	12.1	12.8	11.6	9.7	7.2	5.0	4.4
		MCWBR	4.2	4.5	6.3	7.3	7.3	7.3	6.6	6.4	6.0	5.4	4.5	4.3
(40) Clear-Sky Solar Irradiance	taub		0.321	0.340	0.368	0.407	0.399	0.397	0.414	0.410	0.386	0.359	0.336	0.311
	taud		2.295	2.313	2.266	2.207	2.284	2.316	2.285	2.307	2.357	2.399	2.347	2.290
	Ebn at Noon		617	740	799	815	846	851	826	805	776	712	596	545
	Edh at Noon		59	80	104	126	123	120	122	113	95	73	56	49
(44) All-Sky Solar Radiation	RadAvg		0.52	1.17	2.44	4.07	5.15	5.35	5.04	4.11	2.94	1.59	0.65	0.38
	RadStd		0.07	0.17	0.22	0.46	0.52	0.44	0.61	0.31	0.33	0.18	0.08	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.51	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-143	N/A	N/A

Nomenclature: See separate page